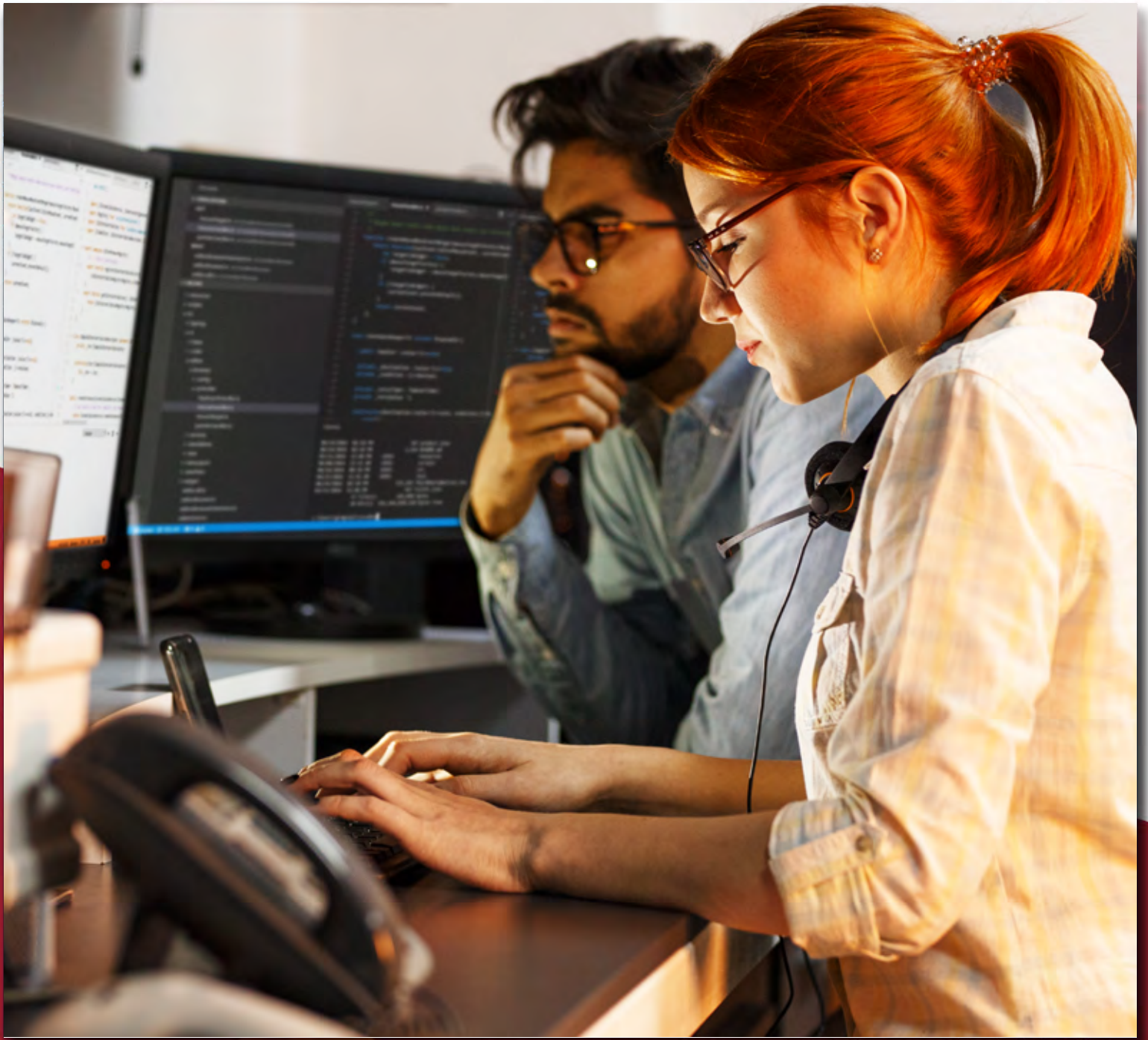




PROFESSIONAL CERTIFICATE IN DATA ENGINEERING

Gain cutting-edge skills to advance your data engineering career.

Delivered in collaboration with



Overview

As the use of technology expands, data engineering is becoming increasingly vital, and the demand for the specialized expertise of data engineers is growing. In fact, according to the Dice Tech Job Report, data engineering was the fastest growing tech occupation in 2020. Why? Because before data scientists can gain useful information from the mountains of data possessed by today's organizations, the data must be configured, warehoused, and made accessible, and data engineers are responsible for building the infrastructure.

The MIT xPRO Professional Certificate in Data Engineering is an immersive 6-month program that's designed to provide you with job-ready, in-demand data engineering skills and a competitive edge in the marketplace. Through an exploration of core concepts, tools, techniques, and best practices, participants will learn data engineering essentials, from building effective data architectures and data warehouses to designing data models, streamlining data processing, automating data pipelines, data wrangling, and big data engineering. Participants also receive personalized feedback, live office hours with learning facilitators, and the opportunity to develop a GitHub portfolio for potential employers.

MIT xPRO's online learning programs showcase industry-aligned content from world-renowned experts to make learning accessible anytime, anywhere, and solve challenges for developing technical professionals.

Take the next step to launch your data engineering career.

Price

USD 7,800

Duration

6 months, online
15-20 hours per week

\$128,631

The average annual pay for a data engineer in the United States.

(Source: Indeed, 2023)

“

“Data engineers build the ‘nervous system’ of the company. Without it, the company cannot react to changes in the external business environment or within the organization. They build the software and hardware systems that power the company's vision and are masters not just of software, but of hardware, networks, and analytic apps that are changing everyday data.”

– John R. Williams, Professor of Information Engineering, MIT

Program Highlights



Earn a certificate and 36 continuing education units (CEUs) from MIT xPRO



Gain insights with coding demos from renowned MIT faculty



Learn market-ready data engineering skills for use in a high-growth market



Build a GitHub portfolio of your projects to share with potential employers



Receive one-on-one career support from Emeritus and introductions to hiring partners for eligible participants



Program Experience



20+ hours
of prerecorded
MIT faculty videos



5+ hours
of live mentorship
and career support*



8 hours
of optional career
development activities*



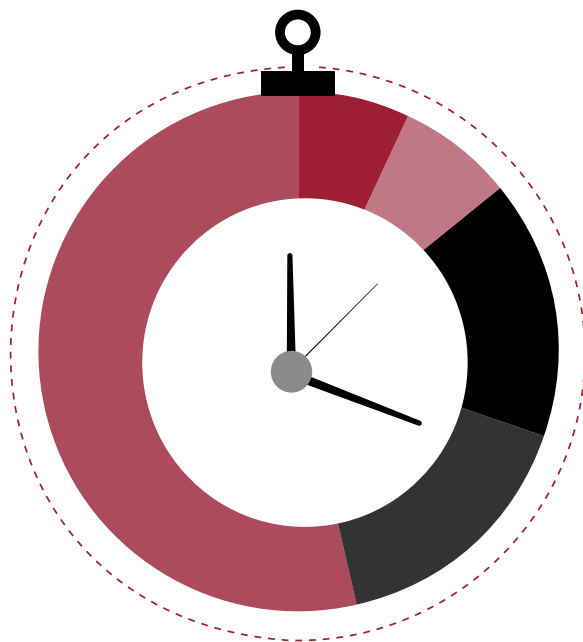
19 career
development video
lectures covering
30 career topics*



Access to résumé
referrals to Emeritus
employer partners
for up to **6 months**
post-program*

Sample Weekly Program Planner

Participants should expect to dedicate a minimum of **15-20** hours per week to the program.



1 hour of recorded video lectures
from faculty



Live interaction with program
learning facilitators**



2 hours of self-study and
practice exercises



2 hours of group discussions with
peers to exchange and generate ideas



13 hours of rigorous, graded assignments
to apply and reinforce the lecture material

*Services provided by Emeritus, a learning partner for this program.

**The schedule for live interactions are subject to change based on availability and will be confirmed once the program starts.

Learning Journey

From navigating the enrollment process to identifying job opportunities, we partner with you to take the next step in your career.

LEARNING FACILITATOR

Your learning facilitator will leverage their industry experience and expertise to guide you by holding live sessions, providing assignment feedback, and answering questions.

LEARNING COMMUNITY

Your learning community will provide an interactive environment where you can learn with a group of like-minded individuals and build a global network of peers.



PROGRAM ADVISOR

Your program advisor will be your enrollment resource, answering any pre-program questions and easing your transition into the program.

CAREER COACH

Your career coach will help you successfully navigate your job search by assisting with goal setting, providing feedback on your cover letter, résumé, and LinkedIn profile, and conducting mock interviews. They will be a source of up-to-the-minute information on hiring trends and help celebrate the next step in your career.

Tools and Resources in the Program

The Professional Certificate in Data Engineering program employs the latest industry tools and resources, including:

- MySQL
- Hadoop
- Docker
- Cassandra
- Google Colab
- Jupyter Notebook
- Kafka
- GitHub
- MySQL
- Spark
- Firebase
- Swagger
- MySQL Workbench
- Visual Studio Code
- MySQL Python Connector Driver
- MySQL Shell
- Git
- Postman
- Strapi
- Flask Web Server
- Java
- Nano Text Editor
- Debezium
- Spring Boot
- Curl
- Mapbox
- Maven
- NiFi
- Wget
- PySpark
- Airflow
- Mosquitto
- ThingsBoard
- Web Sockets
- Node.js
- OAuth2
- Okta
- MongoDB
- Redis
- JSON Web Tokens
- JavaScript
- YAML

Python libraries you will work with:

- NumPy
- pandas
- DASK
- SciPy
- TensorFlow
- Matplotlib
- Seaborn
- Scikit-learn
- OpenSSL
- Gym
- Arrow
- Feather
- Lorem
- Node-rdkafka
- Graphviz
- Data-Driven Documents (D3)
- NotebookJS
- NLTK
- Paho
- Express
- Kafka-Python

"Data engineering really is a core component of today's data infrastructure. And because organizations can't function without data, it's also a career with a great deal of opportunity and incredibly interesting work as well."

– Abel Sanchez, Research Scientist and Executive Director,
MIT's Geospatial Data Center

Who Is This Program For?



Career launchers: Recent STEM graduates/post-graduates/interns looking to start a career in this high-growth field by gaining exposure to data engineering.



Career builders: Early career software engineers/technology professionals seeking training in the latest data engineering tools and techniques to advance their careers.



Career switchers: Mid-career professionals aiming to switch to data engineering from information technology, analytics, finance, project management, supply chain, or other technical fields.

Applicants must have:

- A bachelor's degree or higher
- Strong math skills
- Some programming experience

Also recommended:

- An educational background in STEM fields
- Technical work experience
- Some experience with Python, R, or SQL
- Some experience with statistics and calculus

Prepare for these potential job titles:

- | | | |
|---------------------------|---------------------------|----------------------------------|
| • Data Engineer | • Data Science Engineer | • Data Integration Engineer |
| • Big Data Engineer | • Data Platform Engineer | • Data Infrastructure Engineer |
| • Data Software Engineer | • Python Data Engineer | • Data Systems Engineer |
| • Data Analytics Engineer | • Cloud Data Engineer | • Business Intelligence Engineer |
| | • Data Warehouse Engineer | |

*Participants must be 18 or above to apply for this program.

Key Takeaways

This program is designed to give you the skills you need to start or continue your career in data engineering. High-level learning outcomes for this program include:

- ▶ **Develop and analyze databases using data science and data engineering tools and skills, including SQL and Python**
 - Use multiple Python libraries including NumPy, pandas, DASK, SciPy, TensorFlow, Matplotlib, Seaborn, Scikit-learn, OpenSSL, Gym, Arrow, Feather, Lorem, Node-rdkafka, Graphviz, Data-Driven Documents (D3), NotebookJS, NLTK, Paho, Express, and Kafka-Python
 - Design databases conceptually and formally
 - Perform extract, transform, and load (ETL) on a dataset
 - Perform change data capture (CDC)
 - Develop a web application in Java
 - Connect a database to Debezium
 - Create an application using web tokens
 - Build a transit data application using Mapbox and Maven
 - Use NiFi to create an ETL pipeline
 - Utilize Hadoop to handle big data
 - Use Docker to create and manipulate Spark images and containers
 - Use PySpark to query data
 - Create a workflow in Airflow
 - Learn database containerization, how to use containers when working with databases, and how to run queries to interface with a database container
 - Learn data visualization
- ▶ **Configure a network to ensure data security**
 - Identify the key concepts of security, encryption, and authentication
 - Develop a web application in Java
 - Define web token architecture and create an application using web tokens
- ▶ **Implement artificial intelligence (AI)/machine learning (ML) algorithms, including those for reinforcement learning and deep neural networks**
 - Learn the fundamental concepts of reinforcement learning, including the reward matrix, the quality matrix, the Bellman equation, and deep neural networks
 - Apply gradient descent to reduce error
 - Implement the Naïve Bayes and Gaussian Naïve Bayes theorems and k-means using Scikit-learn
- ▶ **Manage big data using data warehousing and workflow management platforms**
 - Run parallel operations in DASK
 - Stream data through web sockets
 - Identify the key concepts related to visualization, unstructured data, and JavaScript
 - Create a sensemaking data pipeline
- ▶ **Build a user interface to view and interact with large amounts of live streaming data**
 - Discuss use cases for Mosquitto
 - Stream live data to ThingsBoard
 - Analyze live streaming data using ThingsBoard
 - Construct a web server using Kafka
- ▶ **Create a GitHub portfolio to present the projects that you create to potential employers**

Program Learning Objectives

- Explain key data science and data engineering concepts
- Develop and analyze databases using data science and data engineering tools and skills, including SQL and Python
- Configure a network to ensure data security
- Manage big data using data warehousing and workflow management platforms
- Implement AI/ML algorithms, including those for reinforcement learning and deep neural networks
- Build a user interface to view and interact with large amounts of live streaming data

Program Modules

1. Introduction to Python
2. Python: Introduction to NumPy
3. Python: pandas
4. Databases: SQL
5. Databases: Basic SQL Statements
6. Database Analysis and the Client-Server Interface
7. A Model to Predict Housing Prices
8. ETL, Analysis, and Visualization
9. GitHub and Advanced Python Functions
10. Software Engineering Basics
11. Basics of Client-Server Architecture
12. Types of Databases and Database Containerization
13. CDC
14. Java and Debezium
15. Using Advanced Python Programming to Create Web Applications
16. Transit Data and Application Programming Interfaces (APIs)
17. Performing ETL Using NiFi
18. Platforms for Handling Big Data
19. Processing Big Data with Spark and Airflow
20. Introduction to ML and Advanced Probability
21. Introduction to Reinforcement Learning and Deep Neural Networks
22. Processing and Streaming Big Data
23. Creating a Data Pipeline
24. Handling Big Data with Mosquitto, ThingsBoard, and Kafka



Program Schedule

This program is organized into three main sections:

Section 1

- ▶ In the first section of the program, you will learn the basics of the Python programming language, how to work with relational databases using SQL, and how to work with Python to create databases and server pipelines.
- ▶ **Modules 1-3: Python, NumPy, Matplotlib, and pandas**
 - You will work with multiple cutting-edge Python libraries including NumPy, Matplotlib, and pandas. You will use basic data types and advanced structures in Python, such as lists, tuples, sets, and dictionaries.
- ▶ **Modules 4-6: Relational Databases and SQL**
 - You will write complex database queries, use Regular Expressions, clean a database, define drivers to read a table, write files to your database, and write YAML files.
- ▶ **Modules 7 and 8: Portfolio Projects**
 - You will build a prediction modeling using linear regression.
 - You will use ETL to analyze a dataset and then visualize the results using Matplotlib.

Section 2

- ▶ In the second section of the program, you will learn more advanced Python functions and create a GitHub portfolio to present your projects to potential employers. Then, you will dive deeper into various command line and data security tools. You will work on tasks such as database containerization, CDC, and data wrangling.
- ▶ **Modules 9-11: GitHub, Docker, Visual Studio Code, and Flask**
 - You will learn more advanced Python constructs, such as classes, wrappers, and decorators. You will work with a number of software engineering tools, including Postman, Docker, Flask, Bootstrap, cookies, and security tools. You will practice using command line commands, asynchronous event-driven programming, HTTP structure, and creating APIs.
- ▶ **Modules 12-14: Database Containerization, CDC, Java, and Debezium**
 - You will learn how to use containers when working with databases and how to run queries to interface with a database container. You will manipulate data and perform CDC in different types of databases including MongoDB, Cassandra, Redis, and Firebase. You will practice the basics of the Java programming language and use Debezium to perform CDC on containers.
- ▶ **Modules 15 and 16: Portfolio Projects**
 - You will create a web application and use JavaScript Open Notation (JSON) web tokens, authentication, and authorization to create security features. You will also manipulate a database using Python redundant dictionaries.
 - You will use Mapbox and Maven to build a transit data application

Section 3

- ▶ In the third section of the program, you will explore the tools that are used to manage big data and data warehousing. You will learn how ML, reinforcement learning, advanced probability, and deep neural networks are integrated into data engineering.
- ▶ **Modules 17-19**
 - You will use NiFi to construct an ETL pipeline and work with Hadoop, Spark, and Airflow to create data pipelines for big data processing. You will use PySpark to query big data.
- ▶ **Modules 20-22**
 - You will practice foundational ML mathematical algorithms and implement the Python Scikit-learn library. You will stream big data using the pandas, Parquet, and Feather libraries. You will use the DASK library to create, read, write, and analyze multiple files in parallel and simulate parallel processing across distributed machines.
- ▶ **Modules 23 and 24: Portfolio Projects**
 - You will clean data from web pages and use JavaScript, the Document Object Model and HyperText Markup Language to create a sensemaking data pipeline. You will visualize your data with the JavaScript D3 library.
 - You will use an MQTT protocol to produce temperature and humidity data and publish data to ThingsBoard. You will use Kafka to create a Python application that publishes vehicles' location data to a Kafka topic. Finally, you will use Node.js to construct a web server that acts as a consumer for the messages received from the Kafka broker.



Assignments and Portfolio Projects

Each module includes engaging assignments and culminates in at least one GitHub portfolio project that you'll complete based on what you have learned in that portion of the program.

Assignments



Peer discussions



Interactive activities



Practice exercises



Knowledge checks

Coding Exercises

Coding exercises are integrated into various modules through simple activities using Jupyter Notebook. They allow you to practice building composite skills to prepare you for the assignments and portfolio projects.

Portfolio Projects

- Build a predictive ML model involving feature selection for linear regression
- Build a reinforcement learning model for robot navigation (from scratch in Python)
- Run TensorFlow for a deep neural network model (Deep Dream in Colab)
- Build a producer/subscriber broker for visualizing streaming MQTT sensor data
- Stream load 100 million lines of data and create and write 20 files in parallel using DASK
- Protect your web server using JSON web tokens

You will receive personalized feedback from your program leaders on your GitHub repositories, securing a market-ready portfolio that's ready to share with potential employers.

Program Faculty



John R. Williams

Professor of Information Engineering, MIT
Department of Civil and Environmental
Engineering

Dr. John R. Williams holds a B.A. in physics from Oxford University, an M.S. in physics from UCLA, and a Ph.D. in numerical methods from the University of Wales, Swansea. His research focuses on the application of large-scale computation to problems in cyber-physical security and energy studies. He is director of MIT's Geospatial Data Center, and, from 2006–2012, was director of the MIT Auto-ID Laboratory, which invented the Internet of Things (IoT).

Dr. Williams is the author or co-author of over 250 journal and conference papers, as well as the book, *RFID Technology and Applications*. He contributed to the 2013 report for the United Kingdom's Government Office for Science Foresight project – *The Future of Manufacturing*.

Alongside Bill Gates and Larry Ellison, he was named as one of the 50 most powerful people in computer networks. He consults for organizations including Accenture, Schlumberger, SAP Research, Microsoft Research, Kajima Corp, U.S. Lincoln Laboratory, Sandia National Laboratories, U.S. Intelligence Advanced Research Projects Activity, Motorola, Phillip-Morris Inc., Ford Motor Company, ExxonMobil, Shell, Total, and ARAMCO.

Dr. Williams' international collaborations include HKUST and PolyU (Hong Kong), the University of Cambridge and the Imperial College of Science and Technology (United Kingdom), Malaysia University of Science and Technology, KACST (Saudi Arabia), and Masdar Institute of Science and Technology (Abu Dhabi).

He organized the first Cyber-Physical Security Conference in the United Kingdom (2011), and, along with Dr. Sanchez, he runs the MIT Applied Cybersecurity Professional Education summer program. At MIT, he teaches courses in Architecting Software Systems (MIT 1.125) and Engineering Computation and Data Science (MIT 1.00/1.001).

Program Faculty



Abel Sanchez

Research Scientist and Executive Director, the
MIT Geospatial Data Center

Dr. Abel Sanchez holds a Ph.D. from MIT. He is the executive director of MIT's Geospatial Data Center, architect of the IoT global network, and architect of data analytics platforms for SAP, Ford, Johnson & Johnson, Accenture, Shell, ExxonMobil, and Altria. In cybersecurity, Dr. Sanchez architected an impact analysis of large-scale cyber attacks, designing Cyber Ranges for the Department of Defense.

In password security, Dr. Sanchez led the design of a password firewall (negative authentication) for the Intelligence Advanced Research Projects Activity agency. In ML, addressing fraud detection, Dr. Sanchez designed a situational awareness framework that exploits different perspectives of the same data and assigns risk scores to entities for Accenture.

He led the design of a global data infrastructure simulator, modeling follow-the-sun engineering, to evaluate the impact of competing architectures on the performance, availability, and reliability of the system for Ford Motor Company. He is involved in developing e-learning software for Microsoft via their I-Campus Program and establishing the Accenture Technology Academy, an online resource for over 200,000 employees.

He has 10 years of experience with learning management systems that have been deployed in America, Asia, and Europe. He teaches MIT courses on cybersecurity, engineering computation, and data science, and he has produced over 150 educational videos.

Career Preparation and Guidance

This program offers a wide array of career support and guidance to help you develop your career path. These services are provided by Emeritus, our learning collaborator for this program, via the Emeritus Career Center (ECC). The primary goal is to help you build the skills needed to prepare for your career, however we do not guarantee job placement. Learn more about all of the services and support available to you, including:

A SUPPORT TEAM YOU CAN RELY ON

Your support team includes program leaders and career coaches who will help you reach your learning goals and guide you through your job search.

CAREER PREPARATION SERVICES THAT GET YOU NOTICED



Write noteworthy resumes and cover letters



Create effective LinkedIn profiles



Navigate your job search



Prepare for interviews



Craft your elevator pitch



Negotiate your salary



Emeritus Career Center

Comprehensive career services. One convenient location.

The Emeritus Career Center is your one-stop shop for streamlined access to career-related services. As a participant in the program, you will gain lifetime access to the ECC and its related benefits, including:



COACHING

Schedule appointments with a career coach, and share details about new jobs and job search outcomes with them.



EVENT ACCESS

Learn about upcoming career events, register to attend, and view previously recorded webinars.



DOCUMENT STORAGE

Store your resume and other application materials in one convenient location.



RESOURCE LIBRARY

Access our growing Resource Library anytime to find job search resources, resume checklists, and other helpful information.



CAREER PROFILE

Create a profile for networking and to provide valuable information to our employer relations team and potential employers.

These services are provided by Emeritus, our learning collaborator for this program.

UNLOCK ADDITIONAL BENEFITS: SHARE YOUR RESUME

Upload your resume to the ECC for approval and take advantage of:

- A resume review and feedback from your career coach.
- Access to the job board, with postings from our employer partners and others seeking talent.
- An easy application process and the ability to track your application steps.
- Inclusion in resume books that our employer relations team will refer to employers.

Emeritus Hiring Partners

Professional Certificate in Data Engineering is designed to equip you with the in-demand skills that employers seek. The program support team includes dedicated career coaches to guide you through your job search. However, a job placement is not guaranteed.

Over the course of the program, you will gain access to customized recruiting plans developed in collaboration with our hiring partners, connect with hiring managers, obtain personal and industry feedback, and participate in virtual events. Our Employer Partners include:



Our Graduates Work At

Resource
Library

Duke-Energy
Corporation

Ingenovis
Health

Marsh
McLennan

Financing Options

We want to make sure that the Professional Certificate in Data Engineering program is an affordable option for all. This is why we offer you multiple different ways to pay for the program.

Loan Partners (For US Residents)

Climb Credit

Immediate repayment, interest-only repayment, and deferred payment options are available.

- Visit the [Climb Credit application portal](#)
- Fill in your basic details and proceed to the loan section of the application
- Select 'Emeritus/MIT xPRO' under the Campus dropdown, 'Professional Certificate in Data Engineering' from the Program dropdown, and enter your program start date
- Choose your preferred repayment option and enter financial information
- Agree to the disclosure and submit your application
- Our program advisors will contact you for a confirmation on your loan application*
- After confirmation, we will certify your loan. You will receive a welcome email with login instructions from notifications@instructure.com within 3 business days

Sallie Mae

Fixed repayment, interest-only repayment, and deferred payment options are available.

- Visit the [Sallie Mae application portal](#)
- Fill in your basic details and proceed to the loan application page
- Select 'Undergraduate Students' when prompted
- Choose from fixed repayment, interest-only repayment, and deferred payment options, and submit your application
- Our program advisors will contact you for a confirmation on your loan application*
- After confirmation, we will certify your loan. You will receive a welcome email with login instructions from notifications@instructure.com within three business days

Flexible Payment Options (For All Countries)

- Choose to make your payment in two, three, or six installments for higher flexibility.
- Complete your application for the Professional Certificate in Data Engineering, and enroll in the program.

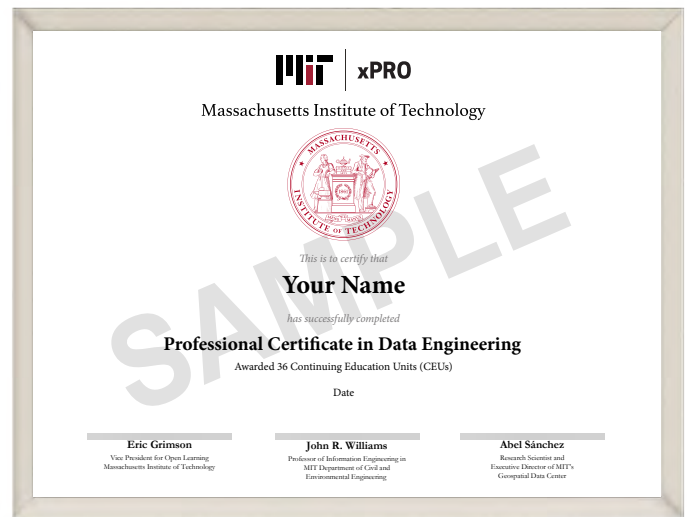
You can opt for any one of the financing options to cover up to the full cost of the program tuition. If you are considering financing your program through one of our partners, the enrollment process can only be completed with the assistance of your program advisor or by calling +1 315 640 4846.

*Due to processing time, the loan application should be submitted no later than four business days prior to the enrollment deadline.

Certificate

Get recognized! Upon successful completion of this program, MIT xPRO grants a certificate of completion to participants and 36 CEUs. This program is graded as a pass or fail; participants must receive 75% to pass and obtain the certificate of completion.

After successful completion of the program, your verified digital certificate will be emailed to you, at no additional cost, with the name you used when registering for the program. All certificate images are for illustrative purposes only and may be subject to change at the discretion of MIT.



About MIT xPRO

MIT xPRO's online learning programs leverage vetted content from world-renowned experts to make learning accessible anytime, anywhere. Designed using cutting-edge research in the neuroscience of learning, MIT xPRO programs are application focused, helping professionals build their skills on the job. To explore the full catalog of MIT xPRO courses and programs, visit: xpro.mit.edu.

About Emeritus

MIT xPRO is collaborating with online education provider Emeritus to deliver this online course through a dynamic, interactive, digital learning platform. This course leverages MIT xPRO's thought leadership in engineering and management practice developed over years of research, teaching, and practice.

You can schedule a call with a program advisor from Emeritus to learn more about this MIT xPRO program.

SCHEDULE A CALL

You can apply to the program here.

APPLY



Refer your colleague and receive a benefit:

REFER NOW

Connect with a program advisor:

Email: learner.success@emeritus.org

Phone: U.S.: +1 315 640 4846

U.K.: +44 141 673 6416

Singapore: +65 3138 2327

Delivered in collaboration with

